

Lab7_Q2: Sierpinski Triangle with Python Turtle

Objective:

Write a Python program using the **TURTLE GRAPHICS** library to draw a **SIERPINSKI TRIANGLE**. This is a fractal made by recursively subdividing an equilateral triangle into smaller triangles.

Problem Description

A Sierpinski Triangle is created by repeatedly drawing smaller triangles within a larger triangle. At each step:

1. Start with a triangle defined by three corner points.
2. Find the **midpoints of each side**.
3. Form **three smaller triangles** using the corners and midpoints.
4. Recursively repeat this process for each smaller triangle until a specified depth.

Input Parameters:

- `length`: The length of the main triangle's side (e.g., 200).
- `depth`: The recursion depth (e.g., 3).

Expected Output:

- A visual representation of a Sierpinski Triangle displayed on the Turtle graphics screen.
- Optionally, you may fill the triangles with color, but this is not required.

Guidelines

1. Implement a **recursive function** that draws a triangle for `depth = 0` and subdivides it for `depth > 0`.
2. Calculate the **midpoints** of each triangle side to determine the smaller triangles.
3. Call the recursive function **three times** for the three new smaller triangles at each step.
4. Use **Turtle graphics** to draw the triangles.
5. Make sure the program runs for various depths (1, 2, 3, etc.) without errors.

Hint

- Each triangle has **three corner points**.
- The midpoint of a line segment between two points $(x1, y1)$ and $(x2, y2)$ can be calculated as:
$$\text{midpoint} = ((x1 + x2)/2, (y1 + y2)/2)$$
- Base case for recursion: when `depth == 0`, draw the triangle and **stop recursion**.
- Recursive case: draw three smaller triangles formed by the corners and midpoints.

